

Identifying and characterizing preventable adverse drug events for prioritizing pharmacist intervention in hospitals

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DOI 10.2146/ajhp160387

Purpose. The approach and results for identifying and characterizing preventable adverse drug events (pADEs) that are key targets for pharmacist medication therapy management (MTM) are summarized.

Methods. This study was part of a larger effort aimed at developing an electronic health record–based prediction model (the complexity score, or C-score) that ranks hospitalized patients according to their risk for pADEs. An environmental scan of published epidemiologic pADE studies and national patient safety priority areas was conducted. The final list of candidate pADEs was then disseminated to ASHP members and a national technical expert panel (TEP) to evaluate the importance, prevalence, severity, preventability, and measurability of these pADEs. Polychoric correlation tests were performed to evaluate and quantify associations between importance and any of the constructs' mean rankings for each individual pADE.

Results. The environmental scan yielded a total of 21 candidate pADEs, including drug-induced acute kidney injury, falls, respiratory depression, altered mental status, hemorrhage, hepatic failure, hypoglycemia, seizures, hypotension and bradycardia, ileus, blood dyscrasias, severe electrolyte imbalances, prolonged hyperglycemia, uncontrolled hypertension, uncontrolled arrhythmia, stress ulcers, hospital-acquired infections, uncontrolled hospital-acquired or community-acquired infection, uncontrolled pain, and venous thromboembolism. The survey confirmed all pADEs were important. Ranking of overall importance was mainly driven by perceived pADE severity and, to a lesser extent, by preventability and prevalence ratings.

Conclusion. A literature review and survey of ASHP and TEP members were used to compile a list of important hospital-acquired pADEs for incorporation into a model for prioritizing patients for MTM services.

Keywords: adverse reactions, prevention and control; drug-related side effects; medication errors, prevention and control; patient safety; pharmacy services, hospital

Am J Health-Syst Pharm. 2017; 74:1774-83

In response to continued patient safety challenges, ASHP proposed the Pharmacy Practice Model Initiative, now termed the Practice Advancement Initiative, which emphasizes pharmacists' key role in optimizing medication-use outcomes.^{1,2} In support of the initiative, ASHP is sponsoring a study to develop a complexity score (C-score) to assist with the allocation of pharmacists to hospitalized

patients with the greatest need for pharmacist services.³ Using fully automated electronic health record (EHR) data, the C-score will be designed to prospectively identify hospitalized patients at high risk for preventable adverse drug events (pADEs).

In developing the C-score, our research group made the decision to target specific pADEs rather than predicting risk for broad adverse out-

comes, such as death or prolonged length of hospital stay. This was motivated by the desire to focus pharmacists' attention on adverse events that are drug related, preventable, and under their direct control. It furthermore allows for the development of explicit and sensitive metrics to capture pharmacists' contributions to patient outcomes. Thus, rather than using a single outcome for prediction, we conceptualized the C-score as an aggregate of several pADEs: the C-score estimates patients' risk for each pADE and then summarizes the aggregate risk across all pADEs into a single score.

In order to prioritize pADEs for inclusion into the C-score, we began with a comprehensive environmental scan of the types of pADEs that have been reported as prevalent or otherwise important in hospitalized patients. The scan included a systematic review of the primary literature as well as publications on pADE-related quality-improvement targets put forward by professional and quality-improvement organizations. While several systematic reviews of pADEs during hospitalization have been published, we noted shortcomings for the purposes of C-score development. Epidemiologic studies have described the frequencies and types of pADEs using a variety of different approaches and classification systems, which complicates attempts to summarize the information. Consequently, several recent reviews did not provide sufficient detail to allow identification of pADE types.⁴⁻⁷ Only 1 prior systematic review of the primary literature published between 1994 and 2001 presented information on the types of pADEs by frequencies of involved drug classes, types of errors that caused the pADEs, and the major organ systems affected.⁸

We presented the results of our environmental scan to pharmacists and medication safety experts to gather insight on the perceived relevance of the identified pADEs for inclusion into the C-score. This article summarizes the approach and results for identifying

KEY POINTS

- A list of prevalent preventable adverse drug events (pADEs) was assembled through a comprehensive literature review.
- ASHP members responding to our survey confirmed that all of the pADEs listed were clinically important focus areas for pharmacist medication therapy management in hospitals.
- Perceived severity, defined as the expected degree of patient harm, was the driving factor when respondents evaluated importance of pADEs.

and characterizing pADEs for C-score development.

Methods

Environmental scan. In order to identify a comprehensive list of candidate pADEs in the hospital setting, we conducted a systematic literature review of studies that reported frequency estimates of types of pADEs. To complement a previously published literature review that included studies published between 1994 and 2001,⁸ we restricted our search to abstracts indexed in PubMed between January 1, 2001, and October 31, 2013, using a combination of search terms that identified the study setting (hospital), the hospital's focus on medication errors, and the presence of an assessment of preventability (appendix). Only publications reporting some quantitative assessments of inpatient pADE or adverse drug event (ADE) frequency by type were included. We excluded studies that exclusively focused on certain specialty populations (e.g., pediatrics), evaluated pADEs or ADEs that caused hospitalization, evaluated the performance of pADE identification tools, or were reviews. Selection of studies was based on titles and abstracts as available in PubMed.

For studies that met inclusion criteria, 1 investigator extracted study characteristics and details on the types of pADEs for which frequency estimates were reported. Because studies reported pADE frequencies by different levels of granularity (e.g., hypotension versus cardiovascular disease), used different pADE ascertainment methods with varying sensitivity, and were conducted in different settings, we did not attempt to extract and summarize pADE frequencies.

We enhanced the information on pADE types from the primary literature by reviewing national quality-improvement targets that address medication-related errors and outcomes. This review included the Healthcare Utilization Project Quality Indicators developed by the Agency for Healthcare Research and Quality,⁹ the core measure sets utilized in the Centers for Medicare and Medicaid Services' Inpatient Quality Reporting Program,¹⁰ and the Joint Commission's National Patient Safety Goals.¹¹ Finally, we conducted a brainstorming session with our health system's clinical pharmacy specialists and medication safety officers to identify pertinent focus areas. The final list of candidate pADEs was then compiled in a survey.

Survey. The survey included all candidate pADEs as well as rating rubrics for overall importance and 4 subconstructs: prevalence, severity, preventability, and measurability. *Overall importance* was defined as respondents' perceptions of each pADE regarding its importance as a component of the C-score and as a factor in prioritizing clinical pharmacist services in hospitalized patients. *Prevalence* was defined as the perceived frequency of each pADE in the hospital setting. Accordingly, *severity* was defined as the expected degree of patient harm associated with each pADE. *Preventability* was defined as the perceived preventability of each pADE through pharmacist intervention, and *measurability* was defined as the ability to measure risk factors prospectively in EHR systems to al-

low the development of an effective prediction model. The final survey instrument asked respondents to supply their professional information; rate the importance, prevalence, severity, preventability, and measurability of each pADE using a 3-point Likert scale; and provide suggestions of additional pADEs not listed. The 3-point Likert scale was defined for each of the 5 sections as follows: importance (1 = not important, 2 = important, and 3 = very important), prevalence (1 = not prevalent, 2 = prevalent, and 3 = very prevalent), severity (1 = not severe, 2 = severe, and 3 = very severe), preventability (1 = not preventable, 2 = preventable, and 3 = very preventable), and measurability (1 = hard to measure, 2 = measurable, and 3 = easy to measure). We integrated survey contents into an online interactive survey platform using Qualtrics software, version 2013 (Qualtrics, Provo, UT). We pilot tested the survey with 20 local clinical pharmacists and pharmacy administrators. Results from the pilot test were not included in the final survey results. The study was approved by the institutional review board of the University of Florida.

We collaborated with ASHP and the ASHP Research and Education Foundation to distribute the survey via the ASHP news link and the ASHP Research and Education Foundation's monthly e-news. The survey link was initially e-mailed to all ASHP members on November 4, 2013, and closed on November 15, 2013. We also distributed the survey to a national technical expert panel (TEP) established to support the C-score project. The TEP included 20 specialists in clinical pharmacy, the development and maintenance of EHR systems, and hospital-based performance and quality measurement.

Results of the survey were expressed as the mean of responses for each pADE by importance, prevalence, severity, preventability, and measurability ratings. Polychoric correlation tests were performed to evaluate and quantify associations

between importance and any of the other 4 constructs' mean rankings for each individual pADE. A p -value <0.05 was considered statistically significant. We used SAS software, version 9.4 (SAS Institute, Cary, NC) for all statistical analyses.

Results

Environmental scan. The keyword search yielded 1,172 PubMed entries, and the titles and abstracts of these entries were reviewed. A total of 135 studies were identified as relevant, and the full-text versions of these were retrieved. A total of 18 studies¹²⁻²⁹ satisfied our inclusion and exclusion criteria, including 5 studies¹⁵⁻¹⁹ in intensive care units and 4 studies confined to the inpatient elderly population.²⁰⁻²³ All considered studies assessed the preventability of ADEs using differing definitions and assessment methods. For example, 3 studies^{13,14,29} and 3 performance measures⁹⁻¹¹ considered ADEs that were associated with "an omission, i.e., conditions for which appropriate therapy was not in place or delayed." One pADE, venous thromboembolism (VTE), was not reported by any of the reviewed studies but was included in national performance measures. We excluded pADEs and ADEs reported in the literature that (1) would unlikely manifest during a hospitalization of average duration and thus would typically not be hospital acquired (e.g., osteoporosis, hypothyroidism, heart failure, adrenal insufficiency), (2) are associated with mild symptoms (e.g., cough, nausea), or (3) are unlikely to be preventable (e.g., chemotherapy-induced vomiting, diarrhea from *Clostridium difficile* infection). The final list included 21 candidate pADEs (Table 1).

Survey. Of the 37,391 ASHP members to whom the survey was sent, 5,809 opened the ASHP monthly e-news and 535 clicked on the survey link. We received responses from 211 pharmacists, and 179 submitted rankings. Most responding pharmacists worked in hospitals (80.1%); a similar proportion identified their primary

workplace as either the community (48.7%) or an academic teaching hospital (41.4%). Approximately one third (36.6%) of the responding hospital pharmacists were clinical pharmacists who provided direct patient care on wards. A majority (62.5%) had at least 5 years of work experience. All 21 TEP members completed the survey.

Both the ASHP member respondents and the TEP rated the mean overall importance of each pADE as greater than 2 on a 3-point Likert scale. Both groups ranked drug-induced hemorrhage and uncontrolled hospital-acquired infections as most important (mean $\pm$ S.D. scores, 2.87 ± 0.37 and 2.78 ± 0.48 for drug-induced hemorrhage and uncontrolled hospital-acquired infections, respectively, as rated by ASHP member respondents; 2.88 ± 0.55 and 2.94 ± 0.24 for drug-induced hemorrhage and uncontrolled hospital-acquired infections, respectively, as rated by TEP members) (Table 2). More than 10% of ASHP member respondents rated drug-induced ileus (10.2%) and new onset of stress ulcers (14.2%) as "not important" pADEs. Because ASHP member survey respondents and TEP members submitted similar rankings, the remaining results are combined for brevity.

Table 3 displays the mean $\pm$ S.D. ratings for prevalence, severity, preventability, and measurability, with pADEs arranged in order of descending mean importance ratings. Severity ratings corresponded closely to overall importance ratings, with mean $\pm$ S.D. scores ranging from 1.72 ± 0.64 (for new onset of stress ulcers) to 2.79 ± 0.44 (for drug-induced hemorrhage). Mean prevalence ratings were consistently lower than their corresponding importance ratings, with mean $\pm$ S.D. scores ranging from 2.41 ± 0.65 (for uncontrolled pain) to 1.26 ± 0.52 (for drug-induced altered mental status). Mean $\pm$ S.D. ratings for preventability ranged from 2.74 ± 0.47 (for prolonged hyperglycemia) to 1.82 ± 0.57 (for uncontrolled arrhythmia). Some discrepancies were noted between sever-

Table 1. Types of pADEs Identified in Environmental Scan^a

pADE(s), by Affected System	Related Concepts for Consideration in Complexity Score
Hepatic	
Liver failure ^{14,15,21,24,29}	Drug-induced hepatic failure
Renal ¹⁸	
Renal failure ^{12-17,19-21,29}	Drug-induced acute kidney injury
Cardiovascular ^{9,18}	
Dysrhythmia (tachycardia, arrhythmia) ^{12-16,21-23,28,29}	Uncontrolled arrhythmia
Hypertension ^{13,14,16,21,22}	Uncontrolled hypertension
Hypotension and bradycardia ^{13,14,16,21-23,25-29}	Drug-induced hypotension and bradycardia
Heart failure ²¹	Not considered
Hematologic ¹⁸	
Bleeding ^{12-14,16,19,20,23,24,26,27,29} or INR increase ^{12,14,21,25}	Drug-induced hemorrhage
Anemia, thrombocytopenia, leukopenia, and neutropenia ^{13-17,19,21,22,24,25,28,29}	Drug-induced blood dyscrasia
Venous thromboembolism ^{9,10}	Venous thromboembolism
Central nervous system ^{4,18}	
Change in mental status ^{12-15,17,21-29}	Drug-induced altered mental status ^b
Seizure ^{12,21-23,29}	Drug-induced seizure
Gastrointestinal ¹³	
Constipation and obstipation ^{14,17,21,22,26,27,29}	Drug-induced ileus
Stress ulcer ^{9,15,21,22,28,29}	New onset of stress ulcer
Diarrhea ^{14,17,21,22,28,29}	Not considered
Nausea and vomiting ^{14,17,21,22,26-29}	Not considered
Metabolic and endocrine	
Hypoglycemia ^{12,21,22,26,27,29}	Drug-induced hypoglycemia
Hyperglycemia ^{9,13-15,21,22,26,29}	Prolonged hyperglycemia
Electrolyte imbalances ^{9,13-16,19-22,26,27,29}	Drug-induced severe electrolyte imbalances ^c
Hypothyroidism ^{21,24,28,29}	Not considered
Adrenal insufficiency ²²	Not considered
Respiratory ¹⁸	
Respiratory depression ^{9,12-15,21,25-27,29}	Drug-induced respiratory depression
Coughing ²²	Not considered
Musculoskeletal	
Fall and fracture ^{11,14,21,26}	Drug-induced falls
Osteoporosis ^{24,29}	Not considered
Other	
Uncontrolled pain ^{13,22,26}	Uncontrolled pain
Onset of infection or delayed recovery from infection ^{11,13,14,19,21,23,24,29}	Manifestation of hospital-acquired infection, uncontrolled hospital-acquired infection, uncontrolled community-acquired infection

^apADE = preventable adverse drug event, INR = International Normalized Ratio.^bIncludes confusion, delirium, hallucination, agitation, dizziness, fatigue, lethargy, and disorientation.^cIncludes hypokalemia, hyperkalemia, and hyponatremia.

Table 2. Overall Importance Ratings Provided by ASHP Member Respondents and TEP Members^a

Type of pADEs	ASHP Members (n = 179)			Mean ± S.D. Rating	TEP Members (n = 21)	
	% Respondents Rating pADE as Very Important	% Respondents Rating pADE as Important	% Respondents Rating pADE as Not Important		Mean ± S.D. Rating From TEP Members (n = 21)	
Drug-induced hemorrhage	88.3	10.6	1.1	2.87 ± 0.37	2.88 ± 0.55	
Uncontrolled hospital-acquired infection	80.7	16.5	2.8	2.78 ± 0.48	2.94 ± 0.24	
Drug-induced respiratory depression	68.4	29.4	2.3	2.66 ± 0.52	2.59 ± 0.50	
Drug-induced acute kidney injury	65.9	33.0	1.1	2.65 ± 0.50	2.53 ± 0.51	
Venous thromboembolism	67.6	30.2	2.2	2.65 ± 0.52	2.65 ± 0.49	
Drug-induced hypoglycemia	66.7	30.5	2.8	2.64 ± 0.54	2.65 ± 0.59	
Drug-induced falls	64.4	32.8	2.8	2.62 ± 0.54	2.53 ± 0.51	
Manifestation of hospital-acquired infection	62.5	34.7	2.8	2.60 ± 0.55	2.47 ± 0.51	
Drug-induced severe electrolyte imbalances	62.1	36.2	1.7	2.60 ± 0.52	2.71 ± 0.46	
Uncontrolled pain	52.5	43.9	4.5	2.48 ± 0.59	2.53 ± 0.51	
Drug-induced altered mental status	48.6	48.0	3.4	2.45 ± 0.63	2.47 ± 0.62	
Drug-induced hepatic failure	52.3	40.3	7.4	2.45 ± 0.63	2.47 ± 0.62	
Uncontrolled community-acquired infection	49.2	45.8	5.1	2.44 ± 0.59	2.47 ± 0.62	
Drug-induced blood dyscrasia	47.1	48.9	4.0	2.43 ± 0.57	2.53 ± 0.62	
Drug-induced seizures	50.6	42.0	7.5	2.43 ± 0.63	2.47 ± 0.62	
Prolonged hyperglycemia	44.6	50.3	5.1	2.40 ± 0.59	2.65 ± 0.59	
Uncontrolled arrhythmia	42.9	49.2	7.9	2.35 ± 0.62	2.41 ± 0.50	
Uncontrolled hypertension	38.6	54.5	6.8	2.32 ± 0.60	2.41 ± 0.51	
Drug-induced hypotension or bradycardia	31.5	61.2	7.3	2.24 ± 0.58	2.18 ± 0.71	
Drug-induced ileus	31.6	58.2	10.2	2.21 ± 0.61	2.24 ± 0.65	
New onset of stress ulcers	26.1	59.7	14.2	2.12 ± 0.63	2.12 ± 0.58	

^aRatings are based on a 3-point Likert scale, where 1 = not important, 2 = important, and 3 = very important. TEP = technical expert panel, pADE = preventable adverse drug event.

Table 3. Prevalence, Severity, Preventability, and Measurability Ratings of pADEs Provided by ASHP Member Respondents (*n* = 179)^a

Type of pADEs (Mean ± S.D. Importance Rating) ^b	Mean ± S.D. Rating		
	Prevalence ^c	Severity ^d	Preventability ^e
Drug-induced hemorrhage (2.87 ± 0.37)	2.11 ± 0.78	2.79 ± 0.44	2.32 ± 0.58
Uncontrolled hospital-acquired infection (2.78 ± 0.48)	1.79 ± 0.73	2.56 ± 0.63	2.35 ± 0.60
Drug-induced respiratory depression (2.66 ± 0.52)	2.04 ± 0.71	2.66 ± 0.49	2.41 ± 0.66
Venous thromboembolism (2.65 ± 0.52)	2.13 ± 0.70	2.37 ± 0.68	2.41 ± 0.56
Drug-induced acute kidney injury (2.65 ± 0.50)	1.99 ± 0.68	2.40 ± 0.61	2.26 ± 0.57
Drug-induced hypoglycemia (2.64 ± 0.54)	2.26 ± 0.63	2.37 ± 0.55	2.51 ± 0.57
Drug-induced falls (2.62 ± 0.54)	2.04 ± 0.72	2.35 ± 0.62	2.25 ± 0.59
Drug-induced severe electrolyte imbalances (2.60 ± 0.52)	2.26 ± 0.63	2.43 ± 0.65	2.34 ± 0.64
Manifestation of hospital-acquired infection (2.60 ± 0.55)	2.01 ± 0.70	2.41 ± 0.60	2.11 ± 0.66
Uncontrolled pain (2.48 ± 0.59)	2.41 ± 0.65	2.17 ± 0.60	2.50 ± 0.58
Drug-induced altered mental status (2.45 ± 0.63)	1.26 ± 0.52	2.44 ± 0.73	2.10 ± 0.73
Drug-induced hepatic failure (2.45 ± 0.63)	2.22 ± 0.69	2.14 ± 0.60	1.96 ± 0.65
Uncontrolled community-acquired infection (2.44 ± 0.59)	1.75 ± 0.67	2.19 ± 0.62	2.08 ± 0.67
Drug-induced seizures (2.43 ± 0.63)	1.64 ± 0.64	2.33 ± 0.66	1.98 ± 0.65
Drug-induced blood dyscrasia (2.43 ± 0.57)	1.33 ± 0.56	2.29 ± 0.64	1.83 ± 0.58
Prolonged hyperglycemia (2.40 ± 0.59)	2.21 ± 0.76	2.03 ± 0.62	2.74 ± 0.47
Uncontrolled arrhythmia (2.35 ± 0.62)	1.66 ± 0.66	2.36 ± 0.64	1.82 ± 0.57
Uncontrolled hypertension (2.32 ± 0.60)	2.11 ± 0.68	2.16 ± 0.65	2.35 ± 0.61
Drug-induced hypotension or bradycardia (2.24 ± 0.58)	1.97 ± 0.66	1.96 ± 0.62	2.02 ± 0.59
Drug-induced ileus (2.21 ± 0.61)	1.83 ± 0.69	1.99 ± 0.65	2.10 ± 0.51
New onset of stress ulcers (2.12 ± 0.63)	1.38 ± 0.58	1.72 ± 0.64	2.39 ± 0.61

^apADE = preventable adverse drug event.^bRatings are based on a 3-point Likert scale, where 1 = not important, 2 = important, and 3 = very important.^cRatings are based on a 3-point Likert scale, where 1 = not prevalent, 2 = prevalent, and 3 = very prevalent.^dRatings are based on a 3-point Likert scale, where 1 = not severe, 2 = severe, and 3 = very severe.^eRatings are based on a 3-point Likert scale, where 1 = not preventable, 2 = preventable, and 3 = very preventable.^fRatings are based on a 3-point Likert scale, where 1 = hard to measure, 2 = measurable, and 3 = easy to measure.

ity and preventability ratings. Several pADEs had mean severity ratings that were at least 0.4 points higher than their corresponding preventability ratings. These included drug-induced hemorrhage, drug-induced hepatic failure, drug-induced blood dyscrasia, and uncontrolled arrhythmia. In contrast, those pADEs having considerably higher preventability than severity ratings were uncontrolled pain, prolonged hyperglycemia, and new onset of stress ulcers. Mean $\pm$ S.D. ratings for measurability were lowest for uncontrolled community-acquired infection (1.71 ± 0.65) and highest for drug-induced severe electrolyte imbalances (2.54 ± 0.58). Measurability scores of <2 on the 3-point Likert

scale indicated perceived difficulties in measurement of clinical manifestations and risk factors using EHR data and included new onset of stress ulcers, drug-induced seizures, drug-induced altered mental status, uncontrolled arrhythmia, and uncontrolled community-acquired infection.

Table 4 shows the correlations between rankings of overall importance and each of the 4 constructs for each individual pADE. Polychoric correlation coefficients range between -1 and 1 ; 1 indicates a total positive correlation, 0 indicates no correlation, and -1 indicates a total negative correlation between 2 constructs. A strong correlation is generally considered with a correlation coefficient of

≥ 0.7 or ≤ -0.7 , and a moderate correlation refers to a correlation coefficient between 0.5 and 0.7 or -0.5 and -0.7 . As expected, rankings of importance showed significant correlations with those of severity across all pADEs. Preventability and importance rankings showed mostly significant but weaker correlations, with polychoric correlation coefficients between 0.251 and 0.497 . Prevalence and measurability were only significantly correlated with approximately half of all ADEs' importance rankings.

Several correlation patterns are noteworthy. First, the mean $\pm$ S.D. importance rankings of drug-induced hemorrhage (2.87 ± 0.37), prolonged hyperglycemia (2.40 ± 0.59), and un-

Table 4. Correlation Between Mean Ratings of Importance of pADEs and Mean Ratings of Prevalence, Severity, Preventability, and Measurability of Those pADEs^a

pADEs in Descending Order of Rated Importance	Polychoric Correlation Coefficients ^b			
	Prevalence	Severity	Preventability	Measurability
Drug-induced hemorrhage	0.124	0.387 ^c	0.107	-0.097
Uncontrolled hospital-acquired infection	0.328 ^c	0.617 ^c	0.362 ^c	0.451 ^c
Drug-induced respiratory depression	0.152 ^c	0.530 ^c	0.239 ^c	0.260
Drug-induced acute kidney injury	0.319 ^c	0.506 ^c	0.383 ^c	0.282 ^c
Venous thromboembolism	-0.007	0.208 ^c	0.335 ^c	0.216
Drug-induced hypoglycemia	0.219	0.631 ^c	0.340 ^c	0.240
Drug-induced falls	0.393 ^c	0.448 ^c	0.251 ^c	0.389 ^c
Manifestation of hospital-acquired infection	0.219 ^c	0.434 ^c	0.183	0.128
Drug-induced severe electrolyte imbalances	0.197 ^c	0.525 ^c	0.362 ^c	0.394 ^c
Uncontrolled pain	0.277 ^c	0.635 ^c	0.111	0.412 ^c
Drug-induced altered mental status	0.296 ^c	0.471 ^c	0.399 ^c	0.221
Drug-induced hepatic failure	0.072	0.366 ^c	0.334 ^c	0.510 ^c
Uncontrolled community-acquired infection	0.158	0.541 ^c	0.415 ^c	0.418 ^c
Drug-induced blood dyscrasia	0.360 ^c	0.556 ^c	0.497 ^c	0.424 ^c
Drug-induced seizures	0.404 ^c	0.534 ^c	0.415 ^c	0.289 ^c
Prolonged hyperglycemia	0.125	0.415 ^c	-0.040	0.133
Uncontrolled arrhythmia	0.227 ^c	0.382 ^c	0.391 ^c	0.289 ^c
Uncontrolled hypertension	0.042	0.459 ^c	0.170	0.187
Drug-induced hypotension or bradycardia	0.154	0.558 ^c	0.252 ^c	0.104
Drug-induced ileus	0.149	0.373 ^c	0.247 ^c	0.268 ^c
New onset of stress ulcers	0.089	0.573 ^c	0.122	0.344 ^c

^apADE = preventable adverse drug event.

^bSee text for interpretation of correlation coefficients. Unless otherwise noted, correlations are not significant.

^c $p < 0.05$.

controlled hypertension (2.32 ± 0.60) were correlated with severity but with no other constructs. This implies that drug-induced hemorrhage was considered the most important pADE because it was thought to carry significantly harmful consequences, but rankings of prevalence and preventability were much lower. Likewise, importance rankings for hyperglycemia and hypoglycemia, which were considerably lower, were driven by lower severity but did not correlate with perceived prevalence and preventability, which were ranked higher. Second, high importance rankings for VTE and drug-induced hypoglycemia were largely driven by high severity and preventability rankings, while prevalence was rated lower and not correlated with the overall importance.

Respondents made several suggestions for additional pADEs, but none were considered for inclusion in C-score development. Suggestions included pADEs that were likely not hospital acquired (e.g., heart failure, chronic obstructive pulmonary disease, asthma, worsening of Parkinson's disease, pancreatitis), were difficult to measure or were rare (allergic reactions, phenytoin or digoxin toxicity, serotonin syndrome, extensive number of days on [medical] device), were largely not preventable (e.g., organ rejection), or were not adverse outcomes but rather errors or evidence of malpractice in processes of care (e.g., drug-drug interactions, multiple medication changes during hospitalization, lack of patient understanding of drug regimen, failure to measure concentrations of drugs that require therapeutic monitoring). Based on the survey rankings and their overwhelming presence in the pADE and quality-improvement literature, all candidate pADEs were considered for use in C-score development.

Discussion

Our study aggregated the major themes of inpatient pADEs via a comprehensive environmental scan that supplemented concepts in the pri-

mary epidemiologic literature with national quality-improvement priorities. To our knowledge, this is the first survey in which pharmacists and other medication safety experts assessed the importance of pADEs for prioritization of pharmacist medication therapy management (MTM). The survey results confirmed the importance of the presented pADEs for inclusion in the C-score from the perspective of pharmacists, the target group of the score. It also helped ensure that the selected set of pADEs is comprehensive in directing pharmacists to patients in greatest need for MTM. In addition to identifying pADEs for inclusion in the C-score, the survey aimed to capture the underlying explanations of importance ratings in terms of prevalence, severity, preventability, and measurability. Several findings emerged from this survey.

First, the survey confirmed the importance of all selected pADEs that arose from our environmental scan. Drug-induced hemorrhage, uncontrolled hospital-acquired infection, respiratory depression, nephrotoxicity, and VTE were rated as the 5 most important pADEs, largely driven by corresponding severity ratings. Insulin, antithrombotics, and opioids have been long-standing concerns for pharmacists and have become the primary target in the national action plan for pADE prevention.³⁰ It is worthwhile to note that hyperglycemia, uncontrolled pain, and VTE were rated as important, which emphasizes the challenge of properly balancing overtreatment and undertreatment with these agents. Interestingly, VTE was not reported in any of the epidemiologic pADE studies, likely due to a limited focus on omission errors and low frequency. Survey ratings corroborated our decision to consider national performance measure priorities in addition to epidemiologic findings to create a comprehensive set of high-priority pADEs.

Second, reported prevalence ratings were generally lower than importance ratings and not necessarily asso-

ciated with each other. ADEs that were considered less prevalent (prevalence mean rating of <2) included drug-induced seizure, drug-induced hepatic failure, uncontrolled arrhythmia, and drug-induced blood dyscrasia. Importance ratings of these pADEs were still above 2, suggesting that prevalence carried less weight in setting the priorities for pharmacist care.

Third, for each pADE, the single strongest correlate of importance ratings was the severity rating. The pADEs with high importance ratings that were correlated with both high severity and high preventability ratings included drug-induced respiratory depression, acute kidney injury, VTE, hypoglycemia, falls, and severe electrolyte imbalances. Notable discrepancies between severity and preventability ratings included assessments of drug-induced hemorrhage, acute liver failure, blood dyscrasia, and arrhythmia, all of which were rated as having considerably higher severity than preventability. The reverse was true for uncontrolled pain, stress ulcers, and hyperglycemia, which were rated higher in terms of preventability than severity. In fact, hyperglycemia, which received a lower mean $\pm$ S.D. severity rating (2.03 ± 0.62) was considered the most preventable ADE (2.74 ± 0.47). Such considerations may inform priorities for pharmacist intervention as they select targets that may be more preventable or that may be associated with the most significant harm, or both.

Fourth, several pADEs, while rated important, received an average preventability score of <2 , including uncontrolled arrhythmia, drug-induced blood dyscrasia, drug-induced hepatic failure, and drug-induced seizure. This assessment may reflect, in part, treatment scenarios where adverse effects are expected and accepted given the expected treatment benefit (e.g., chemotherapy). However, limited published data are available regarding the role of inappropriate therapy in these scenarios and the role of clinical pharmacists in preventing these 4 ADEs

during hospitalizations. Further study is warranted to evaluate the preventability of these ADEs.

Fifth, moderate measurability scores for all pADEs emphasize the need to advance EHR systems from documentation tools to standardized recording systems that allow automated retrieval of risk factors and ADE information for development of clinical alerts and prediction tools.

Finally, respondents added only a small number of pADEs to the proposed list, and most of these described rare events (e.g., serotonin syndrome) or highly specialized fields (e.g., organ rejection). This finding suggests that our survey results encompass the major themes for pharmacist MTM. Because we focused on pADEs that are acquired and become manifest during hospitalization, we excluded readmissions in our analysis. This exclusion notwithstanding, survey respondents echoed the national attention that has been given to readmissions, which constitute not only a core component of current performance assessments but also an increasing focus of pharmacist MTM.³¹

While our survey findings are limited by a low response rate, which may be due in part to a short survey duration (12 days), respondents included largely clinical pharmacists and pharmacy administrators in hospital settings who are in need of tools that enhance systemwide, population-based interventions to reduce pADEs. Our findings were also consistent with assessments by our national TEP of medication safety experts who convened to support C-score development. It is also important to note that this survey focused on respondents' perceptions and did not accurately gauge the actual prevalence, severity, preventability, or measurability of the selected pADEs. Further study on preventability in particular would greatly support quality-improvement initiatives.

We reported the survey rating results using mean instead of median values, as they are more commonly

expressed when reporting Likert scale responses.³² In our study, use of the mean allowed subtler distinctions across pADEs (e.g., given the simple 3-level response scale used, the median would have been 2 in most instances).

We successfully assembled important hospital-acquired pADEs that have been reported in the patient safety literature and were confirmed via a survey of ASHP membership and a national TEP. Candidate pADEs were advanced for the development of EHR-based prediction models with the ultimate goal of developing an automated C-score that can prioritize patients for pharmacist MTM services. Results of this work will be presented in upcoming issues of *AJHP*.

Conclusion

A literature review and survey of ASHP and TEP members were used to compile a list of important hospital-acquired pADEs for incorporation into a model for prioritizing patients for MTM services.

Acknowledgments

ASHP and the ASHP Research and Education Foundation are acknowledged for their support in disseminating to their members the invitations to participate in the survey described herein. The following individuals are acknowledged for serving as technical expert panel members in this study: Alison Apple, D.Ph., M.S.; Mary Blegen, Ph.D., RN; Dale Bratzler, D.O., M.P.H.; Michael Brownlee, Pharm.D., M.S.; Kyle Campbell, Pharm.D., M.S.; Daniel Cobaugh, Pharm.D.; Michael Cohen, B.S.Pharm., M.S., Sc.D.; Debby Cowan, Pharm.D.; Fran Cunningham, Pharm.D.; Robert Feroli, Pharm.D.; Allen Flynn, Pharm.D.; Tejal Gandhi, M.D., M.P.H.; Bruce Gordon, Pharm.D.; Shekhar Mehta, Pharm.D., M.S.; Richard Montgomery, B.S.Pharm., M.B.A.; Steve Pickette, Pharm.D.; Rebecca Prevost, Pharm.D.; Paul Szumita, Pharm.D.; Billy Woodward, B.S.Pharm.; and Lisa Zumberg, Pharm.D.

Disclosures

This study was funded by the ASHP Research and Education Foundation. In his prior capacity as vice president of the ASHP Research and Education Foundation, *AJHP* Editor in Chief Daniel J. Cobaugh served as the project officer for the Drug Thera-

py Management Complexity Score study, which included oversight of the contractor selection process. The authors have declared no other potential conflicts of interest.

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Appendix—Search terms, by concept

Hospital setting (“Hospitals”[mesh] OR “Medication systems, hospital”[mesh] OR “Hospital administration”[mesh] OR “Pharmacy service, hospital”[mesh] OR “Hospital”[All Fields] OR “Hospitals”[All Fields] OR “hospitalized*”[All Fields] OR “hospitalization”[All Fields] OR “Inpatient”[All Fields] OR “Intensive care”[All Fields] OR “ICU”[All Fields] OR “ward”[All Fields] OR “wards”[All Fields])

AND

Medication errors (“adverse drug event*”[All Fields] OR “adverse drug events”[All Fields] OR “adverse drug reaction*”[All Fields] OR “adverse drug reactions”[All Fields] OR “medication error*”[All Fields] OR “medical error”[All Fields] OR “adverse drug effect*” [All Fields] OR “adverse drug effects” [All Fields])

AND

Preventable (“Avoidable” [All Fields] OR “Preventability”[All Fields] OR “Preventable”[All Fields] OR “Prevention”[All Fields])

NOT

“Emergency Service, Hospital”[mesh] OR “Outpatient Clinics, Hospital”[mesh] OR “outpatient”[ti] OR “ambulatory”[ti] OR “emergency”[ti] OR “systematic review*”[tiab] OR “health technology assessment”[tiab] OR “meta analysis”[tiab] OR “meta-analysis”[tiab] OR “review”[Publication Type]